

# What is an Attribute?

## Definition

*An attribute is a property or characteristic of an entity. In a relational database, attributes are represented as columns in a table.*

each attribute has a name , data type , The data type tells the DBMS:

- How much memory to allocate
- What operations are valid
- How the value should be stored

, then the attribute has constraints Constraints enforce business rules.Common constraints:

- NOT NULL
- UNIQUE
- CHECK
- DEFAULT
- PRIMARY KEY
- FOREIGN KEY .

## 1. Simple Attribute

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### Definition

*An attribute that cannot be meaningfully divided into smaller subparts.*

## 2. Composite Attribute

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### Definition

*An attribute that can be logically divided into smaller attributes.*

Example:

Name : aadya rastogi -> aadya + rastogi (fname + lname)

## Single-Valued Attribute

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### Definition

*An attribute that contains exactly one value for each entity.*

## Multi-Valued Attribute

### Definition

*An attribute that may contain multiple values for a single entity.*

Example:

Phone Numbers

Problems with this multi valued attribute:

- Difficult to search
- Difficult to index
- Violates atomicity
- Violates First Normal Form (1NF)
- Hard to update

hence BAD: Suppose we store:

| Student | Phones       |
|---------|--------------|
| Aadya   | 77276, 79870 |

Correct Design

Instead:

#### Students

| StudentID | Name  |
|-----------|-------|
| 101       | Aadya |

#### Phones

| StudentID | Phone      |
|-----------|------------|
| 101       | 9876543210 |
| 101       | 9123456789 |

## Stored Attribute

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### Definition

*An attribute whose value is physically stored in the database.*

## Derived Attribute

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### Definition

*An attribute whose value is computed from other stored attributes.*

*Derived Attribute*

### Definition

*An attribute whose value is computed from other stored attributes.*

## Mandatory Attribute

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Also called NOT NULL attribute.

### Definition:

*Every tuple must contain a value*

Mandatory attributes usually include:

- Primary Key